DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, NANOMINIATURE, SINGLE ROW, PLUG, POLARIZED SHELL, PIN CONTACTS, VERTICAL PCB THROUGH-HOLE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32139.

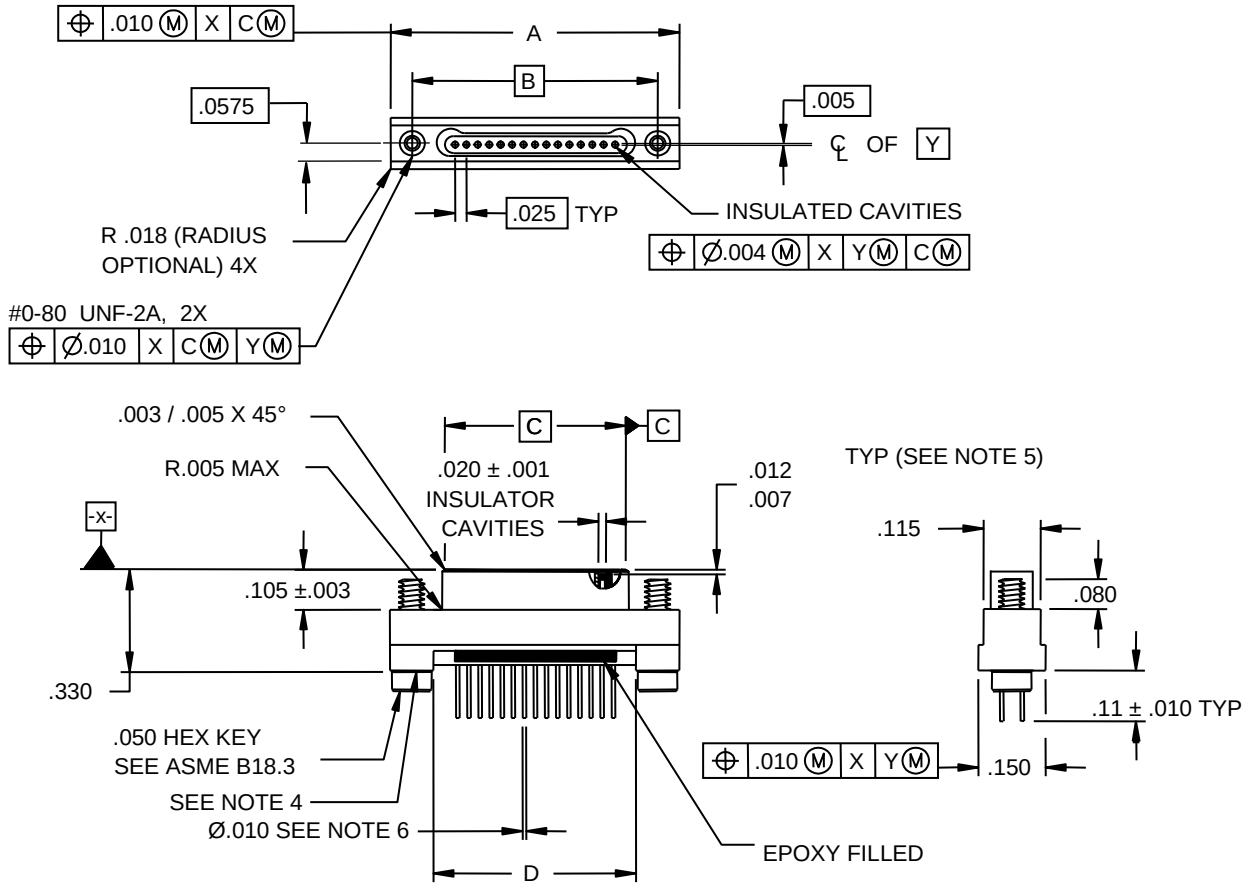


FIGURE 1. Nano connector dimensions and configurations.

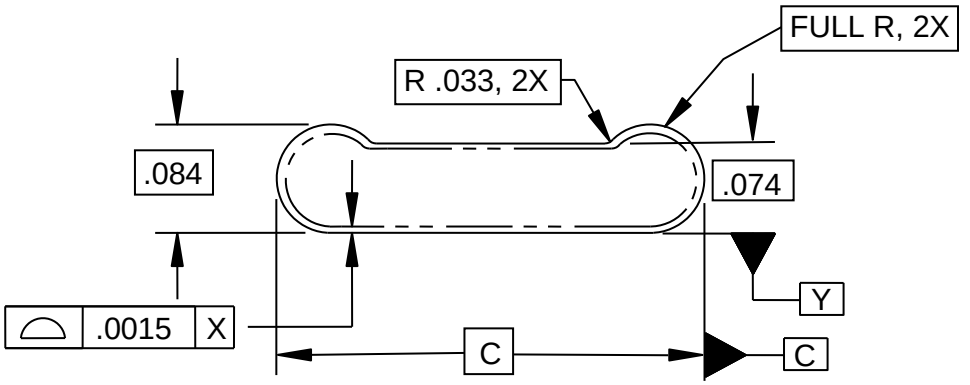
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Insert arrangement	A	B BSC	C BSC	D MAX
9	.500 (12.70)	.395 (10.03)	.284 (7.21)	.295 (7.49)
15	.650 (16.51)	.545 (13.84)	.434 11.02)	.445 (11.30)
21	.800 (20.32)	.695 (17.65)	.584 14.83)	.595 (15.11)
25	.900 (22.86)	.795 (20.19)	.684 17.37)	.695 (17.65)
31	1.050 (26.67)	.945 (24.00)	.834 21.18)	.845 (21.46)
37	1.200 (30.48)	1.095 (27.81)	.984 24.99)	.995 (25.27)
51	1.550 (39.37)	1.445 (36.70)	1.334 33.88)	1.345 (34.16)

Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	0.13
.007	0.18
.010	0.25
.012	0.30
.018	0.46
.020	0.51
.025	0.64
.050	1.27
.0575	1.46
.080	2.03
.105	2.67
.11	2.8
.115	2.92
.150	3.81
.330	8.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) angular tolerance ±2°.
4. Surface from which the lead length is measured.
5. Shell shall be flush to insulator within ± .004 inch (0.10 mm).
6. Tail finish is tin/lead or gold.

FIGURE 1. Nano connector dimensions and configurations – Continued.



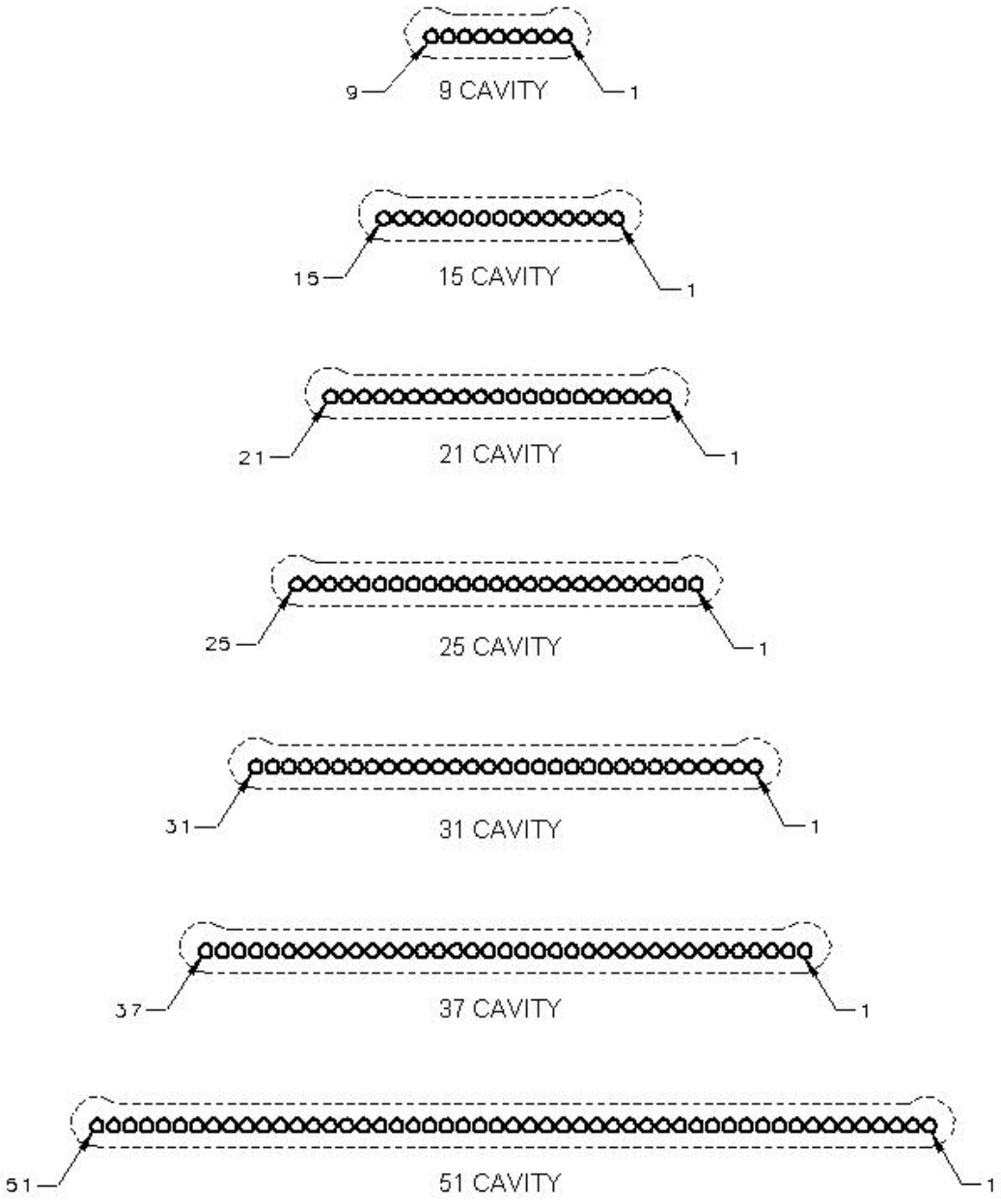
INTERFACE DETAIL

Inches	mm
.0015	0.038
.033	0.84
.074	1.88
.084	2.13

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (.13 mm).

FIGURE 2. Nano connector interface.

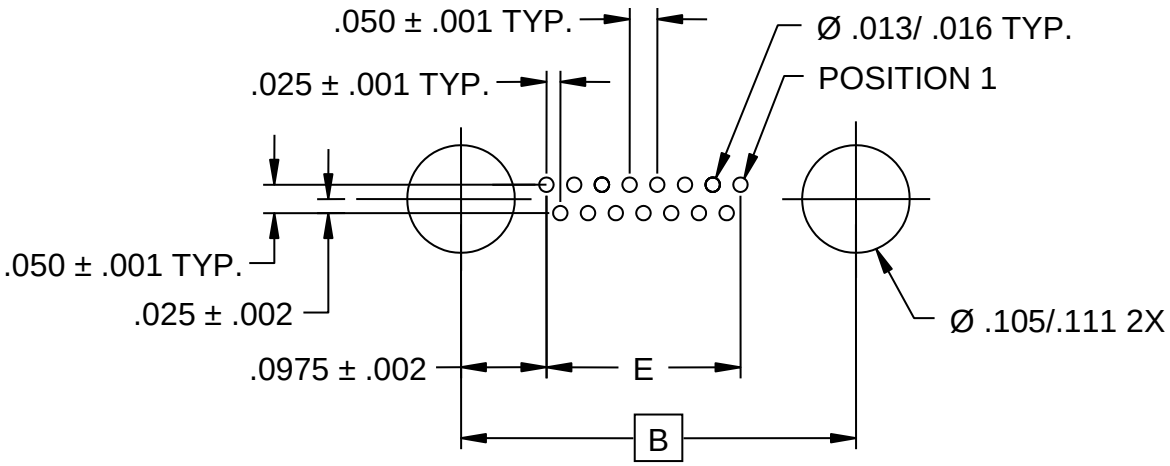
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NOTES:
1. Engaging face of insert shown.
2. Cavity identification numbers are for reference only and do not appear on the part.

FIGURE 3. Nano connector insert arrangement.

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Inches	mm
.001	0.03
.002	0.05
.013	0.33
.016	0.41
.025	0.64
.050	1.27
.0975	2.477
.105	2.67
.111	2.82

Insert arrangement	B BSC	E
9	.395 (10.03)	.200 (5.08)
15	.545 (13.84)	.350 (8.89)
21	.695 (17.65)	.500 (12.70)
25	.795 (20.19)	.600 (15.24)
31	.945 (24.00)	.750 (19.05)
37	1.095 (27.81)	.900 (22.86)
51	1.445 (36.70)	1.250 (31.75)

NOTES:
1. Suggested hole pattern shown facing board from connector insertion side specified on figure 3.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Unless otherwise specified tolerances are ± .005 inch (.13 mm) angular tolerance ±2°.

FIGURE 4. Nano connector suggested PCB hole pattern

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REQUIREMENTS

Dimensions and configuration see figures 1, 2, and 3.

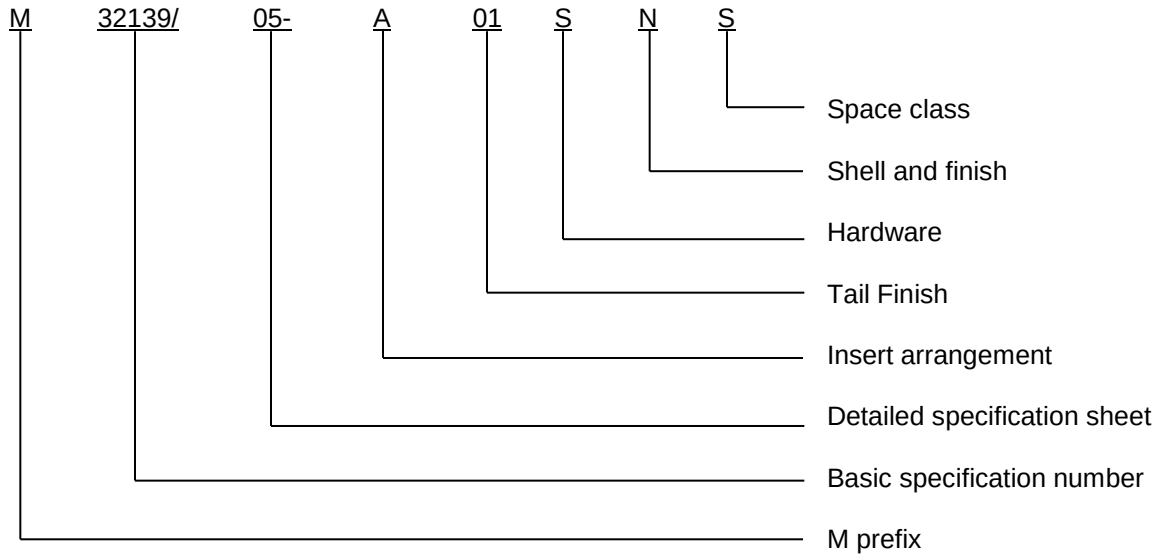
This specification sheet describes the pin side of a rectangular connector. This connector uses reverse gender contacts, i.e., the live pin is recessed in the insulator with the static socket protruding from a shrouded interface.

Contact connection: The pin contact, which is recessed in the insulator, is normally connected to the live side of the circuit.

Pins are terminated to a printed circuit board (PCB) as specified on figure 4.

Mating receptacle: Shall be in accordance with MIL-DTL-32139/6

Part or Identifying Number (PIN):



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Insert Arrangement	Tail Finish
A = 9	01 = Tin/Lead
B = 15	02 = Gold
C = 21	
D = 25	
E = 31	
F = 37	
G = 51	

Hardware [1/]	Shell and Finish	Space class
S = Jackscrew captivated	C = Aluminum shell, cadmium finish	Blank for non-space applications
	N = Aluminum shell, electroless nickel finish [2/]	S = Space class
	S = Passivated stainless steel shell	
	T = Titanium shell	
	A = Pure Electrodeposited Aluminum	
	Z = Zinc Nickel	
	F = Nickel Fluorocarbon Polymer	

1/ Supplied installed.
2/ When aluminum shells are required for space applications, electroless nickel shall be used. Cadmium finish is not acceptable (see MIL-DTL-32139).

Alternate shell finishes:

Pure Electrodeposited Aluminum. Pure dense electrodeposited aluminum shall be in accordance with MIL-DTL-83488, type II shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Nickel Fluorocarbon Polymer. High phosphate nickel with fluorocarbon polymer additive over a suitable underplate shall be able shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Zinc Nickel. Zinc Nickel Alloy shall be in accordance with ASTM B841, over a suitable underplate shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

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Referenced documents. In addition to MIL-DTL-32139, this document references the following:

MIL-DTL-32139/6
MIL-DTL-83488
ASME B18.3
ASTM B841

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AV, MI
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-2010-200)

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